

Bumblebees like flowers...presumably?

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Data Availability: Version-controlled code necessary to reproduce the analyses can be found at <https://github.com/jmelanson98/fvpollens>. Full datasets will be shared to the same repository and archived on Dryad following acceptance.

1 **Abstract**

2 abstract text

3 **Keywords:** keyword1, keyword2, keyword3

4 1 Introduction

5 intro text here

6 2 Methods

7 2.1 Study System

8 2.2 *Bombus* and Floral Surveys

9 2.3 Pollen Metabarcoding

10 2.3.1 Pollen Collection and Quantification

11 For all *B. mixtus* and *B. impatiens* workers, we recorded the presence or absence of cor-
12 bicular pollen as a binary variable. We then removed pollen from the corbiculae of
13 all *B. mixtus* workers and a subset of *B. impatiens* workers; *B. impatiens* workers were
14 selected from a time period (XXX-XXX) to maximize overlap with *B. mixtus* worker
15 activity (MAKE SUPPLEMENTARY FIGURE). This selection of samples allowed us
16 to draw conclusions about season-long and interannual variation in *B. mixtus* pollen
17 collection, while also allowing us to make comparisons between the two species—
18 which have differing phenologies—during a period of temporal overlap when re-
19 source availability was the same for both species.

20 We measured the mass of collected pollen loads using a microbalance (GET BRAND)
21 and stored the pollen at -80°C until DNA extraction. In cases where only a single
22 pollen load was present (e.g., because the second pollen load was dislodge during bee
23 collection) we multiplied the mass of the single pollen load by two to estimate the
24 total mass of pollen collected by the bee.

2.3.2 DNA Extraction, Amplification, and Sequencing

2.3.3 Bioinformatics

We processed ITS2 amplicon sequencing data using the DADA2 pipeline (Callahan et al., 2016) implemented in R version 4.3.3 (R Core Team, 2021) with libraries *dada2* (Callahan et al., 2016), *Biostrings* (Pagès et al., 2026), and *ShortRead* (Morgan et al., 2009). We first trimmed the demultiplexed reads using `truncLen = c(250, 200)`, and then used `cutadapt` (Martin, 2011) to remove primer sequences. We performed a second round of quality filtering to remove low quality reads using the parameters `minLen = c(150,120)`, `maxN= 0`, `maxEE=c(6,8)`, `truncQ= 2`; because *dada2* can canonically handle a large number of errors, a permissive `maxEE` was used at this stage to maintain the maximum number of reads possible for downstream analyses. The DADA2 pipeline estimates the sequence error rates to probabilistically discriminate sequences real sequences from those arising through sequencing errors; we used the `learnErrors` function to generate a parametric model of sequence errors, which was used to perform inference of the assigned sequence variants (ASVs) using the `dada` function (Callahan et al., 2016). Finally, we removed chimeric sequences using the `removeBimeraDenovo` function (Callahan et al., 2016); for all three sequencing runs, the percentage of reads identified as chimeric was between 9-11%

The output of the DADA2 pipeline is a table of ASVs (i.e., unique sequences) and their abundances across samples. We assigned taxonomy to ASVs in QIIME2 (BOYLEN)

2.4 Statistical Analyses

Some ideas for analysis:

- colony-level bumblebee-floral networks (see Ye et al. (2024)) – but find some way to account for phenology

49 3 Results

50 4 Discussion

51 References

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⁷³ **Supporting Information**